Austin Peay State University

PHYS 3051: DC/AC CIRCUITS Lab

Name_________________ Date:__________________

Lab: **Thevenin's and Norton's Theorem**

Purpose: Experimentally determine the Thevenin and Norton equivalent circuits by measuring the open circuit voltage and short circuit currents of the test circuits. Use Thenvenin's and Norton's theorems to reduce complex circuits to simple voltage and current source models. Compute the theoretical equivalents and compare them to the experimental equivalents.

Objectives: Develop circuit construction skills. Develop dc circuit voltage and current measurement skills. Make lab measurements that verify the Thevenin and Norton theorems.

Procedure

1.) Construct the circuit shown in Figure 1 and measure the voltage, V_{ab}, across the 10 kΩ resistor. Record the value of V_{ab} for future use in Table 1.

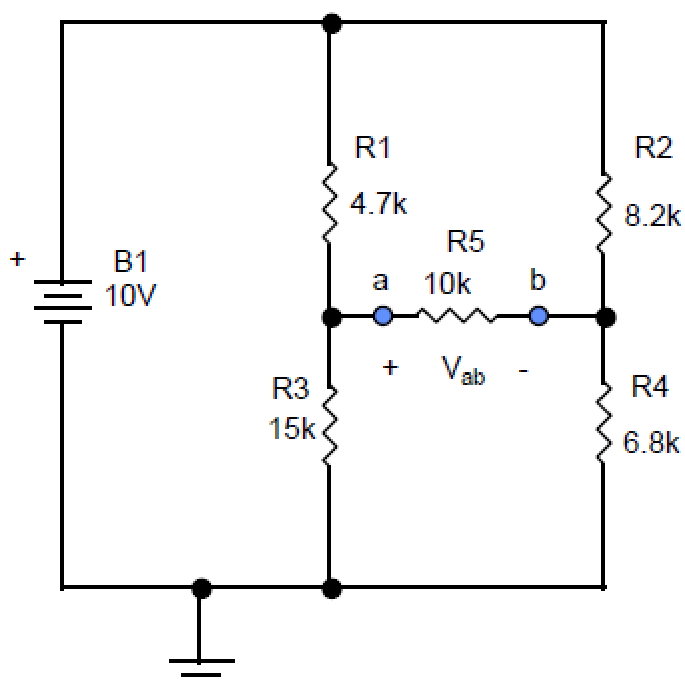

Figure 1. Test Circuit for Part 1of the Experiment.

2.) Remove the 10 kΩ resistor from the circuit in Figure 1 and measure the Thevenin's open circuit voltage, V_{oc}, for the remaining circuit. Figure 2 shows the circuit used to make the open circuit voltage measurements.

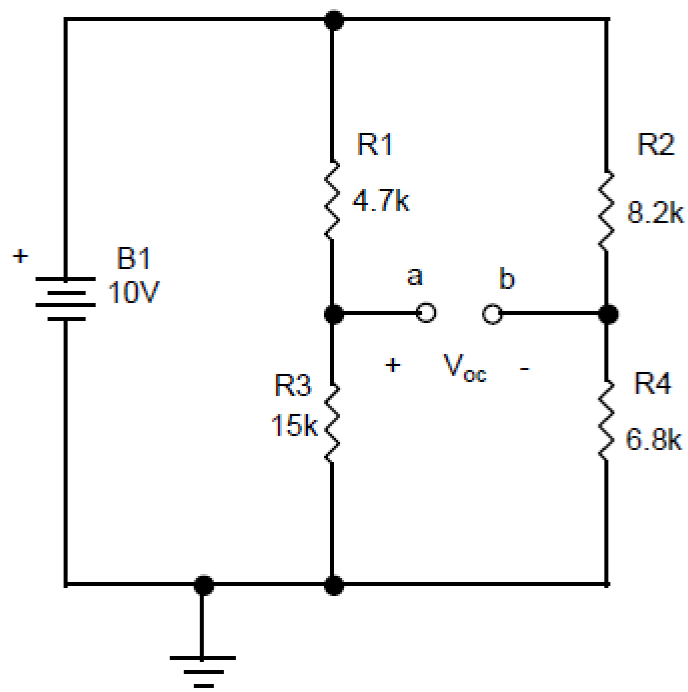

Figure 2. Circuit Connections for Measuring Thevenin's Open Circuit Voltage.

3.) Connect an ammeter between the points a and b as shown in Figure 3 and measure the short circuit current, I_{sc}. Record the value in Table 1 for future use. Note the positive direction of current flow.

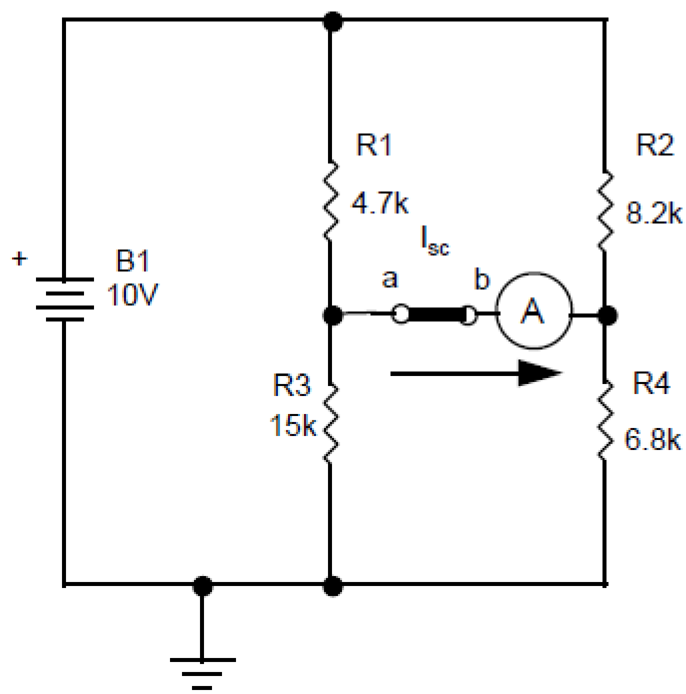

Figure 3. Short Circuit Current Measurement.

4.) From the measurements made in parts 2 and 3, calculate the Thevenin's equivalent resistance for the circuit. This is numerically equal to the Norton's equivalent resistance. Use the formula below to find these values.

$$R_{TH} = R_N = \frac{V_{oc}}{I_{sc}},$$

where: R_{TH} – the Thevenin's equivalent resistance

R_N – the Norton's equivalent resistance

Record these values in Table 1.

5.) Calculate the theoretical values of R_{TH}, R_N, V_{oc}, and I_{sc} for the circuit above. Record the computed values in Table 1.

6.) Using the Thevenin's equivalent circuit, compute the value of V_{ab} with the 10 kΩ resistor attached to points a-b.

7.) Using the Norton's equivalent circuit, compute the value of V_{ab} with the 10 kΩ resistor attached to points a-b.

Report:

1.) Follow the standard laboratory report procedures and format.

2.) Compute the theoretical values for all of the measured superposition values. Include drawings of the Thevenin and Norton equivalent circuits with all values labeled.

3.) Compare the theoretical values to the measured values by computing the percentage error between the theoretical and measured values. Use the formula below to compute the percentage error.

$$\%\text{error} = \frac{(\text{theoretical value - measured value})}{\text{theoretical value}} \times 100\%$$

Discuss the sources of the error.

Table 1- Figure 1 Circuit

Quantity	Value	
	Measured	Theoretical
V_{ab} (V)		
V_{oc} (V)		
I_{sc} (mA)		
R_{TH} (kΩ)		
R_N (kΩ)		
V_{ab} (Thevenin)		
V_{ab} (Norton)		